

Secret Sharing

- Secret sharing is a technique to divide a secret into multiple parts (called *shares*) such that:
 - Only certain groups can reconstruct the secret
 - Others learn **nothing**
- You have a password: “**1234**”
 - You give parts to 3 people such that:
 - Any 2 can recover it; 1 person alone cannot
- This is called a **2-out-of-3 scheme**

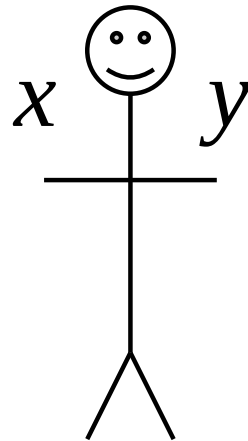
Secret Sharing: “AND” Scheme

- All participants must come together to reconstruct the secret.
- Let $z \in Z_m = \{0, 1, \dots, m-1\}$ be a secret value to be shared with Alice and Bob.
- Randomly and uniformly select values x and y from Z_m subject to the constraint that $(x + y) \bmod m = z$.

Secret Sharing – AND

The secret value is $z = (x + y) \bmod m$.

Me



Let's say you have a secret number z and you want to share it with **Alice and Bob** so that **both must work together** to find it.

Secret Sharing – AND

Example:

- Secret $z=9$, $m=11$
- Choose random $x=4$
- Compute $y=5$

Share:

- Alice $\rightarrow 4$
- Bob $\rightarrow 5$

Reconstruction:

- $(4+5) \bmod 11=9$

Secret Sharing – AND

- This trick easily generalizes to more than two shareholders.
- A secret S can be written as
- $S = (s_1 + s_2 + \dots + s_n) \bmod m$
for any randomly chosen integer values $s_1, s_2, \dots, s_n$ in the range $0 \leq s_i < m$.

Secret Sharing: “OR” Scheme

- I have a secret value z that I would like to share with Alice and Bob such that either Alice or Bob can determine the secret at any time.
- “OR” means **either person alone** can know the secret.
Example: Alice and Bob each get a copy of the secret.
Either one can reveal it.

Example:

Secret = 9

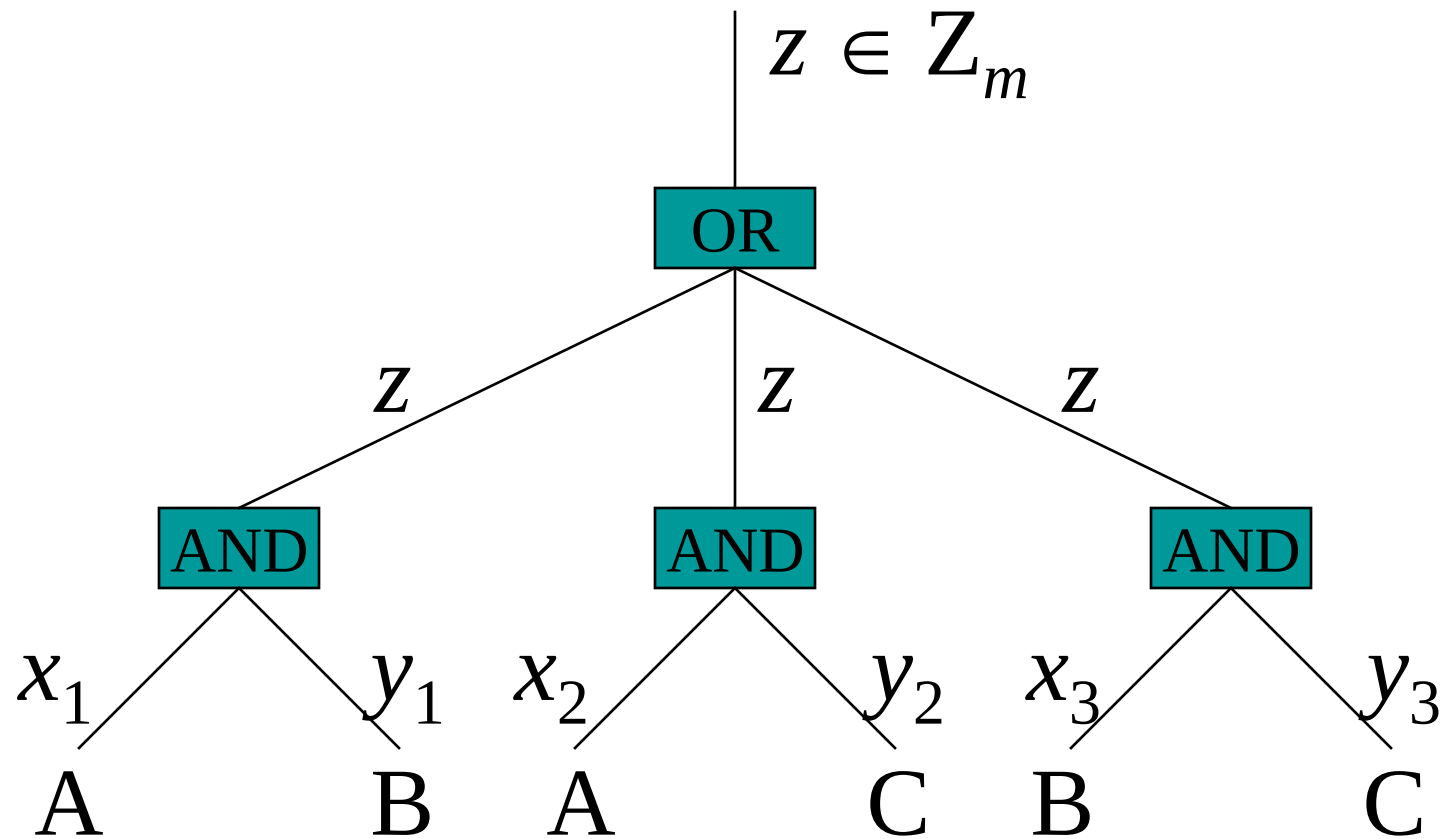
You give: Alice $\rightarrow$ 9 Bob $\rightarrow$ 9

This is not secure sharing, it is just: Replication
(copying)

Secret Sharing Access Structures

- Access Structure is a rule that defines:
Which groups can reconstruct the secret
- We want to share secret value z amongst Alice, Bob, and Carol such that any two of the three can reconstruct z .
- E.g. Allowed groups: $A + B$ $A + C$ $B + C$
 - This can be represented as
$$S = (A \wedge B) \vee (A \wedge C) \vee (B \wedge C)$$

Secret Sharing



Threshold Schemes

A k -out-of- n scheme means:

- Any k people can reconstruct
- Less than $k \rightarrow$ no information

Examples:

2-out-of-3 $\rightarrow$ any 2 needed

3-out-of-5 $\rightarrow$ any 3 needed

Special Cases:

1-out-of- $n \rightarrow$ OR scheme

n -out-of- $n \rightarrow$ AND scheme

Shamir's Threshold Scheme

- Use a polynomial
- Secret = constant term
- Any k points in a field uniquely determine a polynomial of degree at most $k-1$.
- This not only works for the reals, rationals, and other infinite fields, but also over the finite field $Z_p = \{0, 1, \dots, p-1\}$ where p is a prime.

Shamir's Threshold Scheme

Why Modulo (Finite Field)?

Feature 	Without Modulo (Integers/Reals)	With Modulo (Finite Field)
Security	Leaks partial info (e.g., parity)	Perfect Secrecy
Math	Division leads to fractions/errors	Division is always exact
Precision	Floating-point rounding errors	100% exact integer math
Efficiency	Numbers can grow infinitely large	Fixed range $[0, p - 1]$

Shamir's Threshold Scheme

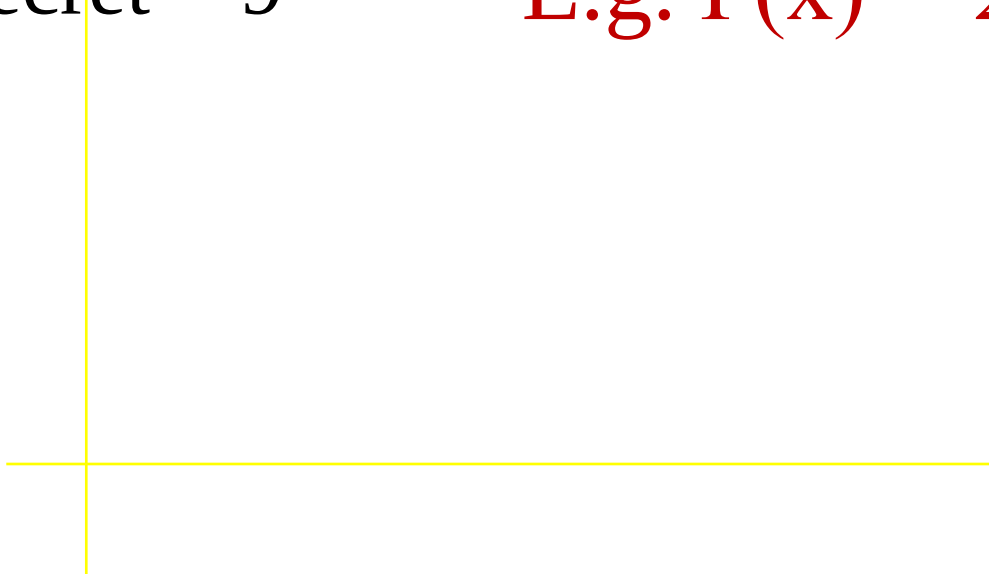
- To distribute a secret value $s \in \mathbb{Z}_p$ amongst a set of n Trustees $\{T_1, T_2, \dots, T_n\}$ such that any k can determine the secret
- pick random coefficients $a_1, a_2, \dots, a_{k-1} \in \mathbb{Z}_p$
- let $P(x) = a_{k-1}x^{k-1} + \dots + a_2x^2 + a_1x + s$
- The secret value is $s = P(0)$.
- give $P(i)$ to trustee T_i . i.e. share $(i, P(i))$

E.g. $P(x) = -2x + 9 \pmod{11}$

For T_1 : $P(1) = -2(1) + 9 = 7$ So, Share = **(1, 7)**

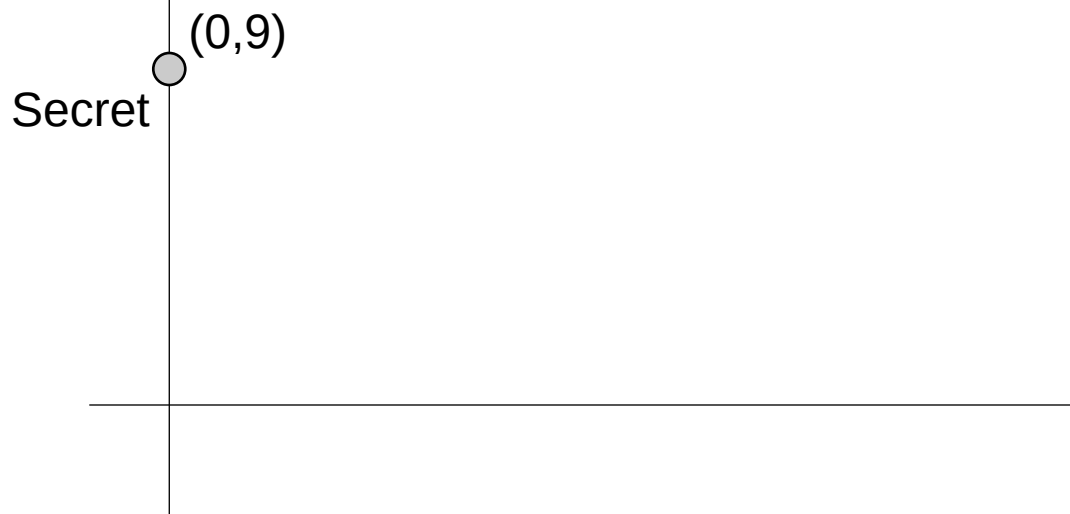
Shamir's Threshold Scheme

- The threshold 2 case:
- Example: Range = $Z_{11} = \{0, 1, \dots, 10\}$,
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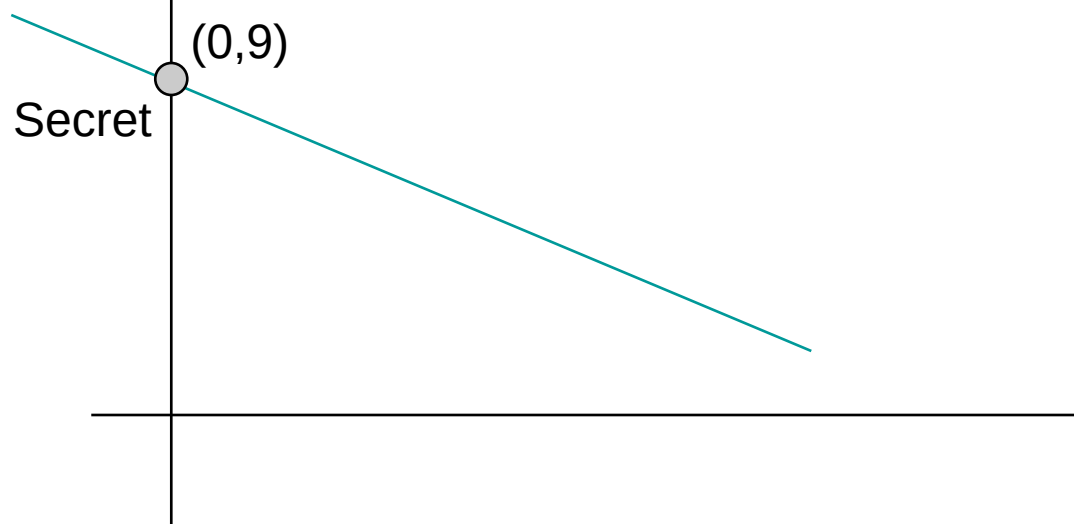
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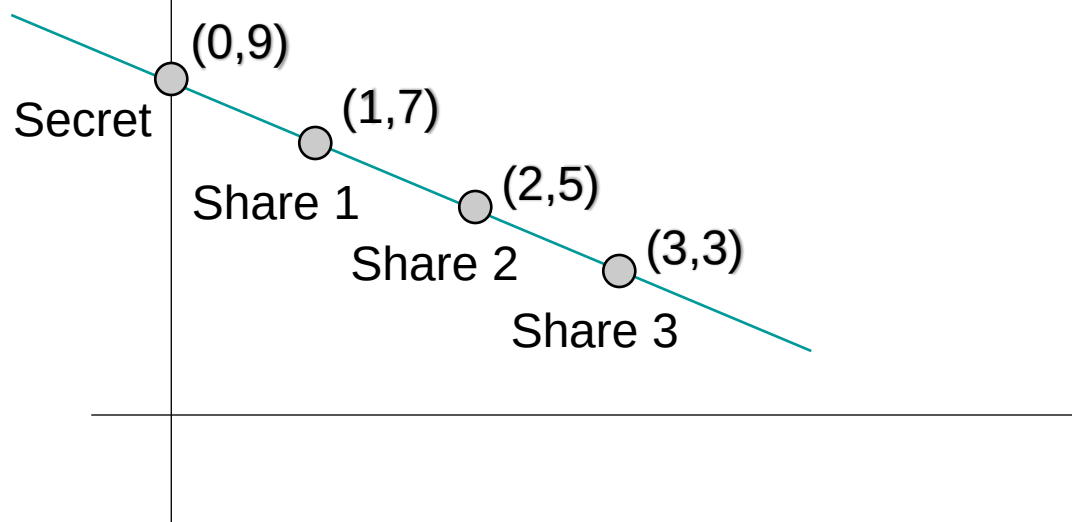
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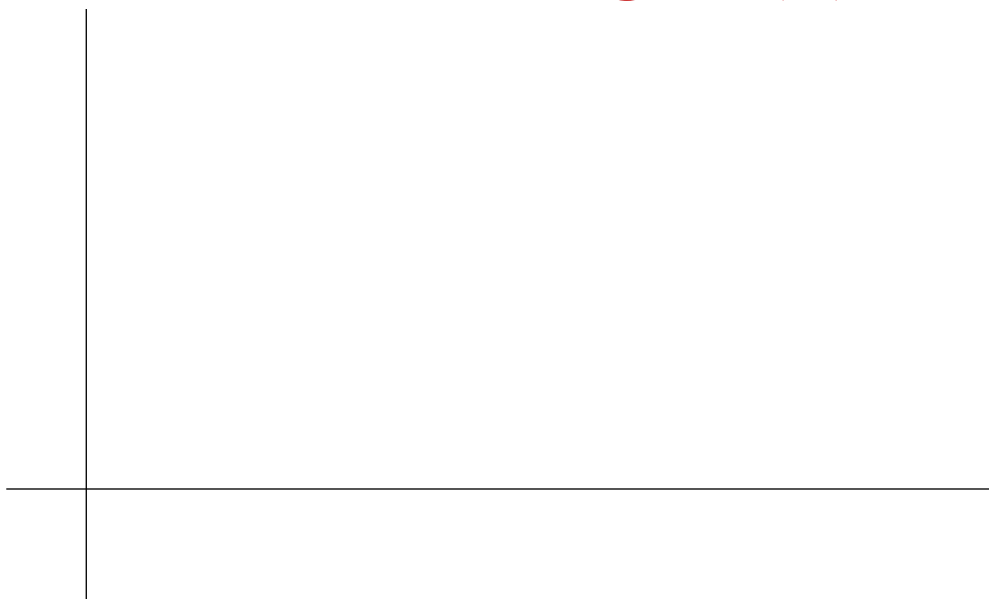
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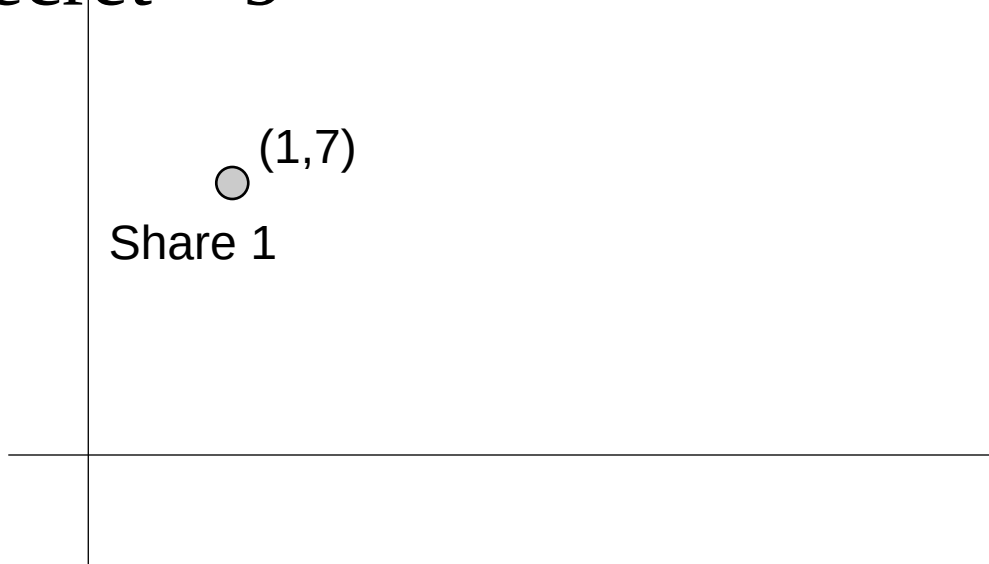
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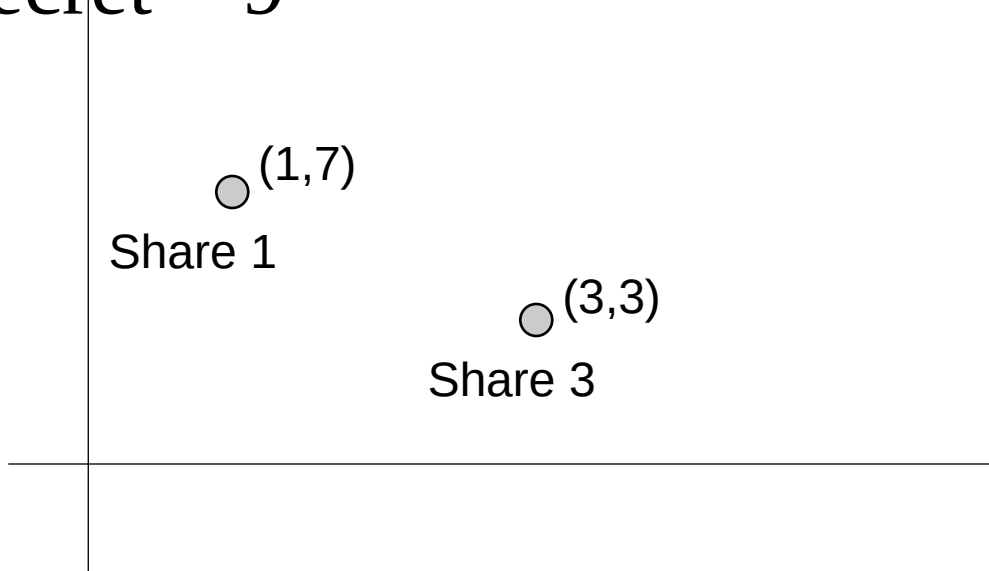
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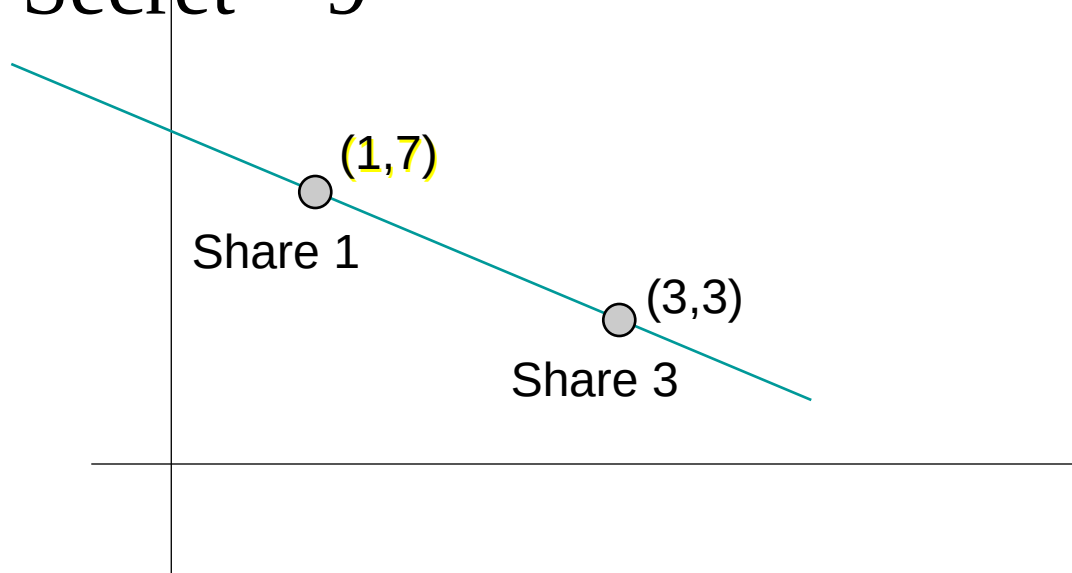
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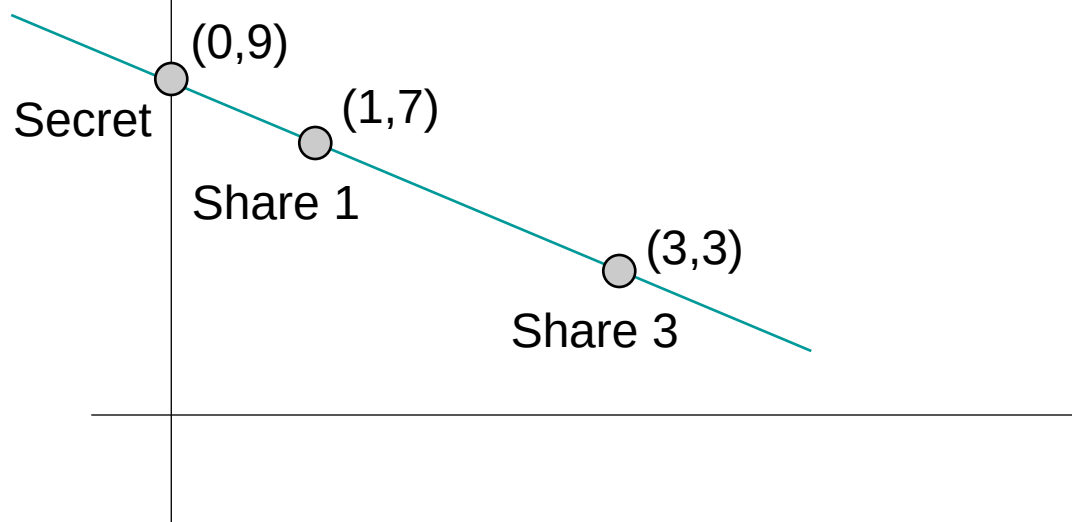
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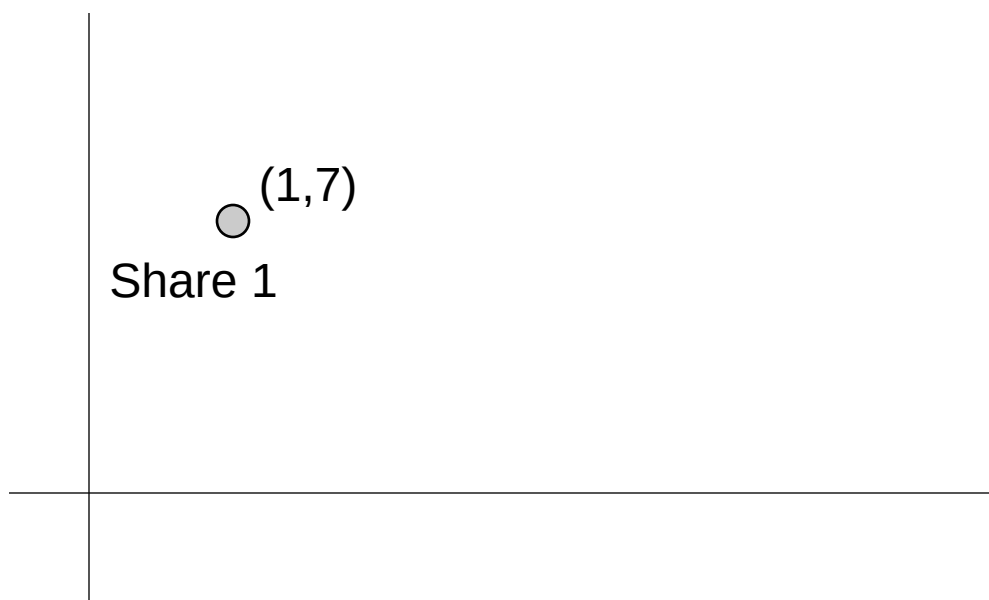
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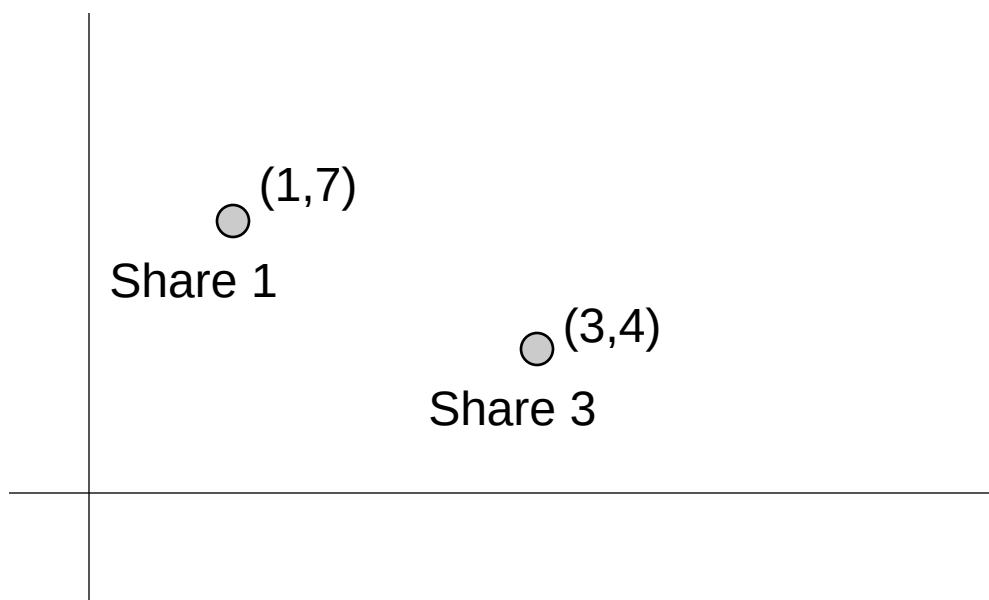
Shamir's Threshold Scheme

Handling decimal problem:



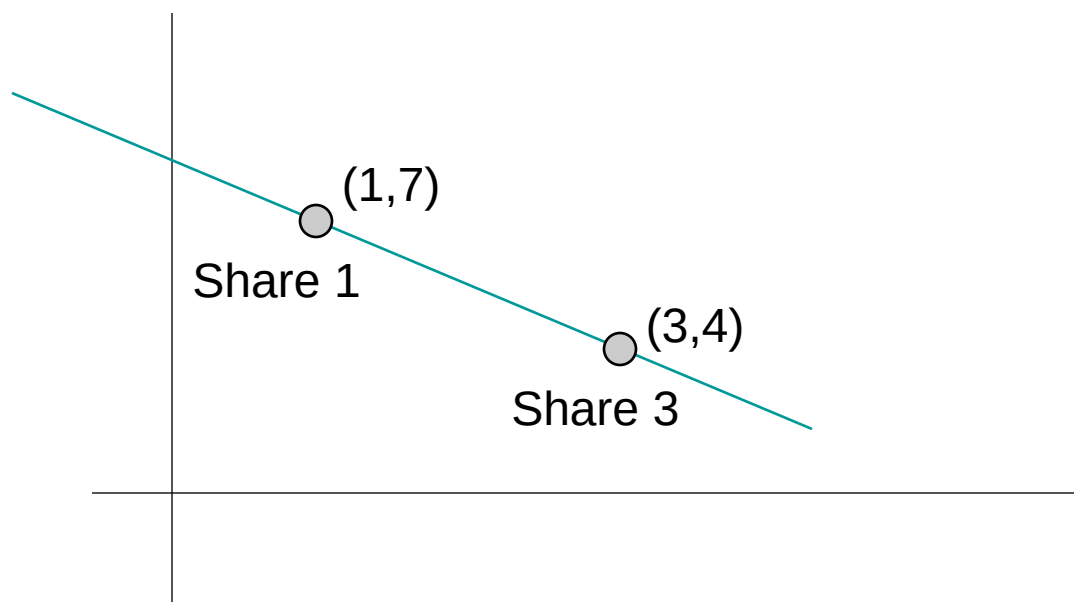
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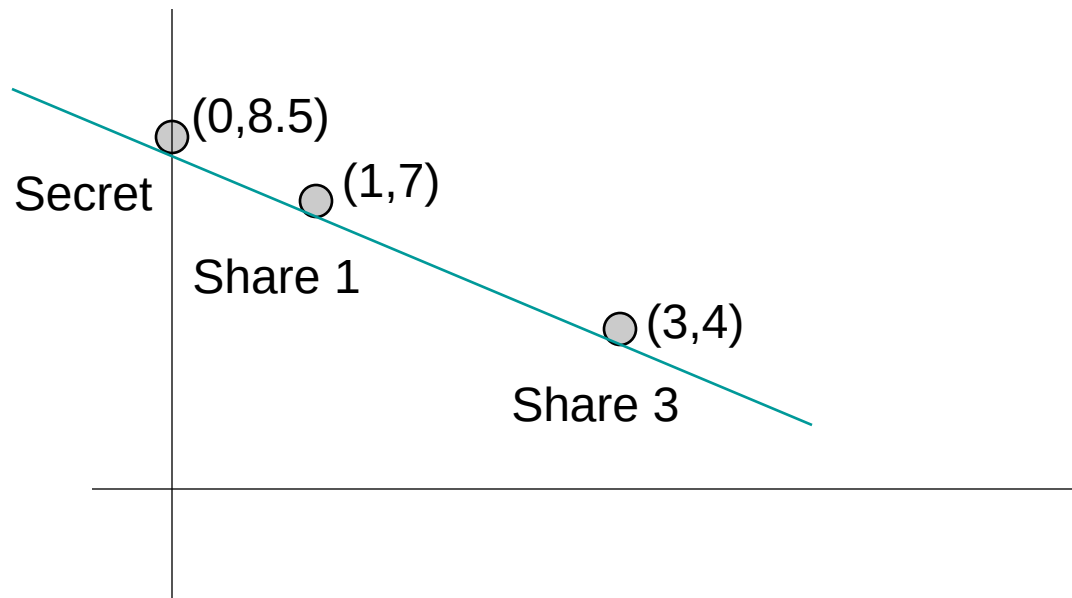


Shamir's Threshold Scheme

What if this is the case:

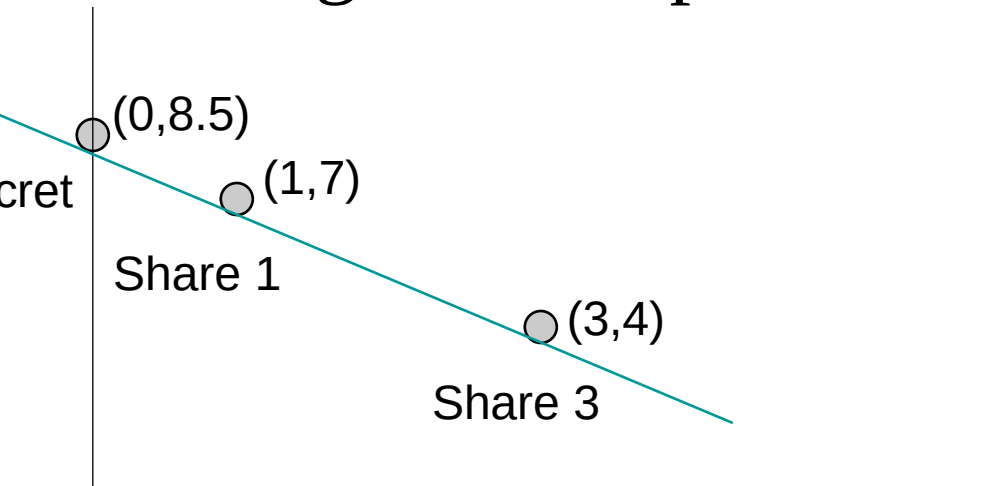


Shamir's Threshold Scheme



Shamir's Threshold Scheme

Handling decimal problem:



By using this method:

- **Decimals are eliminated:** Every value is a whole number.
- **Precision is absolute:** There is no rounding error.
- **Security is maintained:** The secret is a discrete value within the defined field.

$$\begin{aligned} \text{In } Z_{11}, 8.5 &\equiv 17 \div 2 \\ &\equiv 17 \times 2^{-1} \\ &\equiv 17 \times 6 \\ &\equiv 102 \\ &\equiv 3 \end{aligned}$$

Shamir's Threshold Scheme

- Two methods are commonly used to interpolate a polynomial given a set of points:
- Lagrange interpolation
- Solving a system of linear equations

Lagrange Interpolation

- For each point $(i, P(i))$, construct a polynomial P_i with the correct value at i and a value of zero at the other given points.
- $P_i(x) = P(i) \times \prod_{(j \neq i)} (x - j) \div \prod_{(j \neq i)} (i - j)$
- $P(x) = \sum_i P_i(x)$

Lagrange Interpolation

- Generate polynomial with n points
(For degree $n-1$)

$$\begin{aligned} & \frac{(x - x_2)(x - x_3) \dots (x - x_n)}{(x_1 - x_2)(x_1 - x_3) \dots (x_1 - x_n)} * y_1 + \\ & \frac{(x - x_1)(x - x_3) \dots (x - x_n)}{(x_2 - x_1)(x_2 - x_3) \dots (x_2 - x_n)} * y_2 + \dots \\ & \frac{(x - x_1)(x - x_2) \dots (x - x_{n-1})}{(x_n - x_1)(x_n - x_2) \dots (x_n - x_{n-1})} * y_n \end{aligned}$$

General rule for a (t, n) Shamir's Threshold Scheme

Threshold (t)	Type of Equation	Degree	Example Form
2 people	Linear	1	$y = a_1 x + a_0$ (Line)
3 people	Quadratic	2	$y = a_2 x^2 + a_1 x + a_0$ (Parabola)
4 people	Cubic	3	$y = a_3 x^3 + a_2 x^2 + a_1 x + a_0$ (Cubic Curve)
5 people	Quartic	4	

Exercise

- Construct and find the secret value from the following points where the threshold value is 2.
- $(1,3), (2,1), (3, -1)$

Hint: Find $f(x)$ and then find the value $f(x=0)$.

As $k=2$, we can use only 2 points

Using Lagrange interpolation with points $(1, 3)$ and $(2, 1)$:

$$\begin{aligned} f(x) &= 3 \cdot \frac{x-2}{1-2} + 1 \cdot \frac{x-1}{2-1} \\ &= 3 \cdot \frac{x-2}{-1} + (x-1) = -3(x-2) + (x-1) \\ &= -3x + 6 + x - 1 = -2x + 5 \\ f(0) &= 5 \end{aligned}$$

Verification using $(3, -1)$:

$$f(3) = -2(3) + 5 = -6 + 5 = -1$$

Secret = 5

Exercise: Quadratic Example (t=3)

Calculate the hidden secret which uses a quadratic polynomial. You have 3 shares: (1, 8) (2, 7) and (3, 10)

we first find the complete function $f(x)$ and then evaluate it at $f(0)$.

Points: (1, 8), (2, 7), (3, 10)

Step 1: Construct the polynomial $f(x) = 8 * [(x - 2)(x - 3)] / [(1 - 2)(1 - 3)] + 7 * [(x - 1)(x - 3)] / [(2 - 1)(2 - 3)] + 10 * [(x - 1)(x - 2)] / [(3 - 1)(3 - 2)]$

Step 2: Simplify the denominators $f(x) = 8 * [(x^2 - 5x + 6) / 2] + 7 * [(x^2 - 4x + 3) / -1] + 10 * [(x^2 - 3x + 2) / 2]$

Step 3: Distribute the values $f(x) = 4(x^2 - 5x + 6) - 7(x^2 - 4x + 3) + 5(x^2 - 3x + 2)$
 $f(x) = (4x^2 - 20x + 24) - (7x^2 - 28x + 21) + (5x^2 - 15x + 10)$

Step 4: Combine like terms $f(x) = (4 - 7 + 5)x^2 + (-20 + 28 - 15)x + (24 - 21 + 10)$
 $f(x) = 2x^2 - 7x + 13$

Step 5: Find the Secret $f(0) = 2(0)^2 - 7(0) + 13 = 13$

Secret = 13

Exercise: Cubic Example ($t=4$)

Calculate the hidden secret using a cubic polynomial. You have 4 shares: (1, 2), (2, 1), (3, 4), (4, 11)